l7vsdbfir

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[56]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

0.1 1. Load data

1. Load provided `Iris.csv` file.
2. Change the Species columns to numeric label `0, 1, 2` and output a numpy.ndarray in shape `(num_data, 1)`.
3. Convert `SepalLengthCm`, `SepalWidthCm` column into numpy.ndarray in shape `(num_data, 2)`.

1.1 Load provided `Iris.csv` file. Show the first 10 lines (1 points.)

[57]:
```
#########################
# Write your code here #
data = pd.read_csv('Iris.csv')

data.head(10)
#########################
```

[57]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
5	6	5.4	3.9	1.7	0.4	Iris-setosa
6	7	4.6	3.4	1.4	0.3	Iris-setosa
7	8	5.0	3.4	1.5	0.2	Iris-setosa
8	9	4.4	2.9	1.4	0.2	Iris-setosa
9	10	4.9	3.1	1.5	0.1	Iris-setosa

1.2 Change the Species columns to numeric label `0, 1, 2` and output a numpy.ndarray in shape `(num_data, 1)`. Show data every 10 lines. (2 points)

[58]:
```
#########################
# Write your code here #
```

```
species_mapping = {'Iris-setosa': 0, 'Iris-versicolor': 1, 'Iris-virginica': 2}
data['Species'] = data['Species'].map(species_mapping)
data[::10]
########################
```

[58]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	0
10	11	5.4	3.7	1.5	0.2	0
20	21	5.4	3.4	1.7	0.2	0
30	31	4.8	3.1	1.6	0.2	0
40	41	5.0	3.5	1.3	0.3	0
50	51	7.0	3.2	4.7	1.4	1
60	61	5.0	2.0	3.5	1.0	1
70	71	5.9	3.2	4.8	1.8	1
80	81	5.5	2.4	3.8	1.1	1
90	91	5.5	2.6	4.4	1.2	1
100	101	6.3	3.3	6.0	2.5	2
110	111	6.5	3.2	5.1	2.0	2
120	121	6.9	3.2	5.7	2.3	2
130	131	7.4	2.8	6.1	1.9	2
140	141	6.7	3.1	5.6	2.4	2

1.3 Convert `SepalLengthCm` and `SepalWidthCm` columns to ndarray in shape `[num_data, 2]`. Convert numeric label column to ndarray in shape `[num_data, 1]`. Print these two ndarray (1 point).

[59]:
```
########################
# Write your code here #
sepal_array = data[['SepalLengthCm', 'SepalWidthCm']].values
species_array = data['Species'].values.reshape(-1, 1)
y = species_array
print(sepal_array)
print(species_array)

########################
```

```
[[5.1 3.5]
 [4.9 3. ]
 [4.7 3.2]
 [4.6 3.1]
 [5.  3.6]
 [5.4 3.9]
 [4.6 3.4]
 [5.  3.4]
 [4.4 2.9]
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 [4.8 3.4]
 [4.8 3. ]
```

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[4.3 3. ]
[5.8 4. ]
[5.7 4.4]
[5.4 3.9]
[5.1 3.5]
[5.7 3.8]
[5.1 3.8]
[5.4 3.4]
[5.1 3.7]
[4.6 3.6]
[5.1 3.3]
[4.8 3.4]
[5.  3. ]
[5.  3.4]
[5.2 3.5]
[5.2 3.4]
[4.7 3.2]
[4.8 3.1]
[5.4 3.4]
[5.2 4.1]
[5.5 4.2]
[4.9 3.1]
[5.  3.2]
[5.5 3.5]
[4.9 3.1]
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[5.1 3.4]
[5.  3.5]
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[4.4 3.2]
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[5.  2. ]
```

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[5.9 3. ]
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[6.7 3.1]
[5.6 3. ]
[5.8 2.7]
[6.2 2.2]
[5.6 2.5]
[5.9 3.2]
[6.1 2.8]
[6.3 2.5]
[6.1 2.8]
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[6.8 2.8]
[6.7 3. ]
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[5.7 2.6]
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[5.5 2.4]
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[6.  2.7]
[5.4 3. ]
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[6.3 2.3]
[5.6 3. ]
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[5.5 2.6]
[6.1 3. ]
[5.8 2.6]
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 [7.2 3.6]
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 [6.  3. ]
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 [6.7 3. ]
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 [6.5 3. ]
 [6.2 3.4]
 [5.9 3. ]]
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[1]
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[2]
[2]
[2]

0.2 2. Softmax, Cost, and Derivative Functions

2.1 Develop softmax function and try to output softmax function output of this example (3 points). Softmax function

$$\hat{y}_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}, \text{ where } c \text{ is class index and } C \text{ is the number of classes.}$$

[60]:
```
def softmax(z):
    """
    Calculating softmax function.
    Args:
        z: (numpy.ndarray in shape [num_data, num_classes]) logit
    Returns:
        pred: (numpy.ndarray in shape [num_data, num_classes]) predict value
    """
    ##########################
    # Write your code here #
    z_exp = np.exp(z - np.max(z, axis=1, keepdims=True))

    pred = z_exp / np.sum(z_exp, axis=1, keepdims=True)
    ##########################

    return pred

test_z = np.array([[1, 1, 1],
                   [1, 3, 5],
                   [-1, 0, 1]])

softmax(test_z)
```

[60]:
```
array([[0.33333333, 0.33333333, 0.33333333],
       [0.01587624, 0.11731043, 0.86681333],
       [0.09003057, 0.24472847, 0.66524096]])
```

2.2 Develop cost function of multi-class logistic regression and try to output cost function output of this example (3 points). Cost function

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c=1}^{C} y_{ic} \log(\hat{y}_{ic}), \text{ where } N \text{ is the number of data}$$

(Hint: convert label ndarray to the same shape of predicted ndarray, e.g.

$$\begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

[61]:
```
def cost_function(pred, label):
    """
    Calculating cost function.
    Args:
        pred: (numpy.ndarray in shape [num_data, num_classes]) predict value␣
 ↪after softmax
    Returns:
        loss: (numpy.ndarray)
    """
    ########################
    # Write your code here #
    num_samples = label.shape[0]
    num_classes = pred.shape[1]
    one_hot_label = np.zeros((num_samples, num_classes))
    one_hot_label[np.arange(num_samples), label.flatten()] = 1

    loss = -np.sum(one_hot_label * np.log(pred)) / num_samples
    loss = float(loss)
    ########################
    return loss

test_pred = np.array([[0.1, 0.2, 0.7],
                      [0.2, 0.6, 0.2],
                      [0.2, 0.1, 0.7]])
test_label = np.array([[2], [1], [0]])
cost_function(test_pred, test_label)
```

[61]:
```
0.8256461600462744
```

2.3 Develop gradient calculation of multi-class logistic regression and try to output gradient of given example. You're required to derive the derivatives $\frac{\partial \mathcal{L}}{\partial \mathbf{w}}$ by yourself and write it in your writing problems. Assume all $x_0 = 1$, predicted values should be $\hat{y}_i = \text{Softmax}\left(\mathbf{x}_i^T \mathbf{w}\right)$ (3 points for coding, 3 points for mathematical derivation).

[62]:
```
def get_gradient(X, y, y_pred):
    """
    Calculate gradient for gradient descent.
    Args:
        X: (numpy.ndarray in shape [num_data, 3])
        y: (numpy.ndarray in shape [num_data, 1])
        y_pred: (numpy.ndarray in shape [num_data, num_classes])
    Return:
        gradient: (numpy.ndarray in [3, 3]) gradient of w
    """
    ########################
    # Write your code here #
    num_samples = y.shape[0]
```

```
    num_classes = y_pred.shape[1]
    one_hot_y = np.zeros((num_samples, num_classes))
    one_hot_y[np.arange(num_samples), y.flatten()] = 1


    gradient = np.dot(X.T, (y_pred - one_hot_y)) / num_samples
    ##########################
    return gradient

test_X = np.array([[1, 1, 2],
                   [1, 2, 3],
                   [1, 3, 4],
                   [1, 4, 5]])
test_y = np.array([[0], [1], [1], [2]])
test_y_pred = np.array([[0.1, 0.2, 0.7],
                        [0.2, 0.4, 0.4],
                        [0.2, 0.3, 0.5],
                        [0.3, 0.4, 0.3]])
get_gradient(test_X, test_y, test_y_pred)
```

[62]:
```
array([[-0.05 , -0.175,  0.225],
       [ 0.325, -0.375,  0.05 ],
       [ 0.275, -0.55 ,  0.275]])
```

0.3 3. Gradient Descent

3.1 Convert data according to assumption of $x_0 = 1$. (1 point)

[63]:
```
##########################
# Write your code here #
def convert_data_with_x0(X):
    num_samples = X.shape[0]

    x0 = np.ones((num_samples, 1))

    data_with_x0 = np.hstack((x0, X))
    return data_with_x0
X= convert_data_with_x0(sepal_array)
X
##########################
```

[63]:
```
array([[1. , 5.1, 3.5],
       [1. , 4.9, 3. ],
       [1. , 4.7, 3.2],
       [1. , 4.6, 3.1],
       [1. , 5. , 3.6],
       [1. , 5.4, 3.9],
```

```
[1. , 4.6, 3.4],
[1. , 5. , 3.4],
[1. , 4.4, 2.9],
[1. , 4.9, 3.1],
[1. , 5.4, 3.7],
[1. , 4.8, 3.4],
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[1. , 4.3, 3. ],
[1. , 5.8, 4. ],
[1. , 5.7, 4.4],
[1. , 5.4, 3.9],
[1. , 5.1, 3.5],
[1. , 5.7, 3.8],
[1. , 5.1, 3.8],
[1. , 5.4, 3.4],
[1. , 5.1, 3.7],
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[1. , 5.2, 3.4],
[1. , 4.7, 3.2],
[1. , 4.8, 3.1],
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[1. , 5.5, 3.5],
[1. , 4.9, 3.1],
[1. , 4.4, 3. ],
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[1. , 5. , 3.5],
[1. , 5.1, 3.8],
[1. , 4.8, 3. ],
[1. , 5.1, 3.8],
[1. , 4.6, 3.2],
[1. , 5.3, 3.7],
[1. , 5. , 3.3],
[1. , 7. , 3.2],
[1. , 6.4, 3.2],
[1. , 6.9, 3.1],
```

```
[1. , 5.5, 2.3],
[1. , 6.5, 2.8],
[1. , 5.7, 2.8],
[1. , 6.3, 3.3],
[1. , 4.9, 2.4],
[1. , 6.6, 2.9],
[1. , 5.2, 2.7],
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[1. , 6.1, 2.9],
[1. , 5.6, 2.9],
[1. , 6.7, 3.1],
[1. , 5.6, 3. ],
[1. , 5.8, 2.7],
[1. , 6.2, 2.2],
[1. , 5.6, 2.5],
[1. , 5.9, 3.2],
[1. , 6.1, 2.8],
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[1. , 6.4, 2.9],
[1. , 6.6, 3. ],
[1. , 6.8, 2.8],
[1. , 6.7, 3. ],
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[1. , 5.8, 2.7],
[1. , 6. , 2.7],
[1. , 5.4, 3. ],
[1. , 6. , 3.4],
[1. , 6.7, 3.1],
[1. , 6.3, 2.3],
[1. , 5.6, 3. ],
[1. , 5.5, 2.5],
[1. , 5.5, 2.6],
[1. , 6.1, 3. ],
[1. , 5.8, 2.6],
[1. , 5. , 2.3],
[1. , 5.6, 2.7],
[1. , 5.7, 3. ],
[1. , 5.7, 2.9],
[1. , 6.2, 2.9],
[1. , 5.1, 2.5],
[1. , 5.7, 2.8],
```

```
[1. , 6.3, 3.3],
[1. , 5.8, 2.7],
[1. , 7.1, 3. ],
[1. , 6.3, 2.9],
[1. , 6.5, 3. ],
[1. , 7.6, 3. ],
[1. , 4.9, 2.5],
[1. , 7.3, 2.9],
[1. , 6.7, 2.5],
[1. , 7.2, 3.6],
[1. , 6.5, 3.2],
[1. , 6.4, 2.7],
[1. , 6.8, 3. ],
[1. , 5.7, 2.5],
[1. , 5.8, 2.8],
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[1. , 6.9, 3.2],
[1. , 5.6, 2.8],
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[1. , 6.2, 2.8],
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[1. , 6.4, 2.8],
[1. , 7.2, 3. ],
[1. , 7.4, 2.8],
[1. , 7.9, 3.8],
[1. , 6.4, 2.8],
[1. , 6.3, 2.8],
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[1. , 6.9, 3.1],
[1. , 6.7, 3.1],
[1. , 6.9, 3.1],
[1. , 5.8, 2.7],
[1. , 6.8, 3.2],
[1. , 6.7, 3.3],
[1. , 6.7, 3. ],
[1. , 6.3, 2.5],
```

```
       [1. , 6.5, 3. ],
       [1. , 6.2, 3.4],
       [1. , 5.9, 3. ]])
```

3.2 Initialize value of **w** and start up gradient descent. Record cost function value iteration by iteration. (3 points)

[64]:
```python
def gradient_descent(X, y, lr, epoch):
    np.random.seed(42)
    w = np.random.rand(3, 3)

    ########################
    # Write your code here #
    global cost_record
    cost_record = []


    for i in range(epoch):

        logits = np.dot(X, w)
        y_pred = softmax(logits)

        loss = cost_function(y_pred, y)

        gradient = get_gradient(X, y, y_pred)

        w -= lr * gradient

        cost_record.append(loss)


        print(f"[Epoch {i+1}] Cost function = {loss:.4f}")

    return w

    ########################

    return w

epoch = 10000
lr = 0.1
cost_record = []
w = gradient_descent(X, y, lr, epoch)
```

```
[Epoch 1] Cost function = 1.4265
[Epoch 2] Cost function = 1.3192
[Epoch 3] Cost function = 1.2950
[Epoch 4] Cost function = 1.2800
```

```
[Epoch 5] Cost function = 1.2671
[Epoch 6] Cost function = 1.2550
[Epoch 7] Cost function = 1.2433
[Epoch 8] Cost function = 1.2318
[Epoch 9] Cost function = 1.2205
[Epoch 10] Cost function = 1.2095
[Epoch 11] Cost function = 1.1987
[Epoch 12] Cost function = 1.1881
[Epoch 13] Cost function = 1.1777
[Epoch 14] Cost function = 1.1675
[Epoch 15] Cost function = 1.1575
[Epoch 16] Cost function = 1.1476
[Epoch 17] Cost function = 1.1380
[Epoch 18] Cost function = 1.1286
[Epoch 19] Cost function = 1.1194
[Epoch 20] Cost function = 1.1103
[Epoch 21] Cost function = 1.1015
[Epoch 22] Cost function = 1.0928
[Epoch 23] Cost function = 1.0843
[Epoch 24] Cost function = 1.0760
[Epoch 25] Cost function = 1.0678
[Epoch 26] Cost function = 1.0598
[Epoch 27] Cost function = 1.0520
[Epoch 28] Cost function = 1.0444
[Epoch 29] Cost function = 1.0369
[Epoch 30] Cost function = 1.0295
[Epoch 31] Cost function = 1.0223
[Epoch 32] Cost function = 1.0153
[Epoch 33] Cost function = 1.0084
[Epoch 34] Cost function = 1.0017
[Epoch 35] Cost function = 0.9951
[Epoch 36] Cost function = 0.9886
[Epoch 37] Cost function = 0.9823
[Epoch 38] Cost function = 0.9761
[Epoch 39] Cost function = 0.9700
[Epoch 40] Cost function = 0.9641
[Epoch 41] Cost function = 0.9582
[Epoch 42] Cost function = 0.9525
[Epoch 43] Cost function = 0.9470
[Epoch 44] Cost function = 0.9415
[Epoch 45] Cost function = 0.9361
[Epoch 46] Cost function = 0.9309
[Epoch 47] Cost function = 0.9257
[Epoch 48] Cost function = 0.9207
[Epoch 49] Cost function = 0.9158
[Epoch 50] Cost function = 0.9109
[Epoch 51] Cost function = 0.9062
[Epoch 52] Cost function = 0.9015
```

```
[Epoch 53] Cost function = 0.8970
[Epoch 54] Cost function = 0.8925
[Epoch 55] Cost function = 0.8881
[Epoch 56] Cost function = 0.8838
[Epoch 57] Cost function = 0.8796
[Epoch 58] Cost function = 0.8754
[Epoch 59] Cost function = 0.8714
[Epoch 60] Cost function = 0.8674
[Epoch 61] Cost function = 0.8635
[Epoch 62] Cost function = 0.8597
[Epoch 63] Cost function = 0.8559
[Epoch 64] Cost function = 0.8522
[Epoch 65] Cost function = 0.8486
[Epoch 66] Cost function = 0.8450
[Epoch 67] Cost function = 0.8415
[Epoch 68] Cost function = 0.8381
[Epoch 69] Cost function = 0.8348
[Epoch 70] Cost function = 0.8314
[Epoch 71] Cost function = 0.8282
[Epoch 72] Cost function = 0.8250
[Epoch 73] Cost function = 0.8219
[Epoch 74] Cost function = 0.8188
[Epoch 75] Cost function = 0.8158
[Epoch 76] Cost function = 0.8128
[Epoch 77] Cost function = 0.8098
[Epoch 78] Cost function = 0.8070
[Epoch 79] Cost function = 0.8041
[Epoch 80] Cost function = 0.8014
[Epoch 81] Cost function = 0.7986
[Epoch 82] Cost function = 0.7959
[Epoch 83] Cost function = 0.7933
[Epoch 84] Cost function = 0.7907
[Epoch 85] Cost function = 0.7881
[Epoch 86] Cost function = 0.7856
[Epoch 87] Cost function = 0.7831
[Epoch 88] Cost function = 0.7807
[Epoch 89] Cost function = 0.7783
[Epoch 90] Cost function = 0.7759
[Epoch 91] Cost function = 0.7736
[Epoch 92] Cost function = 0.7713
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3.3 Plot cost function v.s. epoch curve. (1 point)

[65]:
```
########################
# Write your code here #
epochs = range(epoch)


plt.figure(figsize=(8, 6))
plt.plot(epochs, cost_record, label="Cost Function", color="blue")

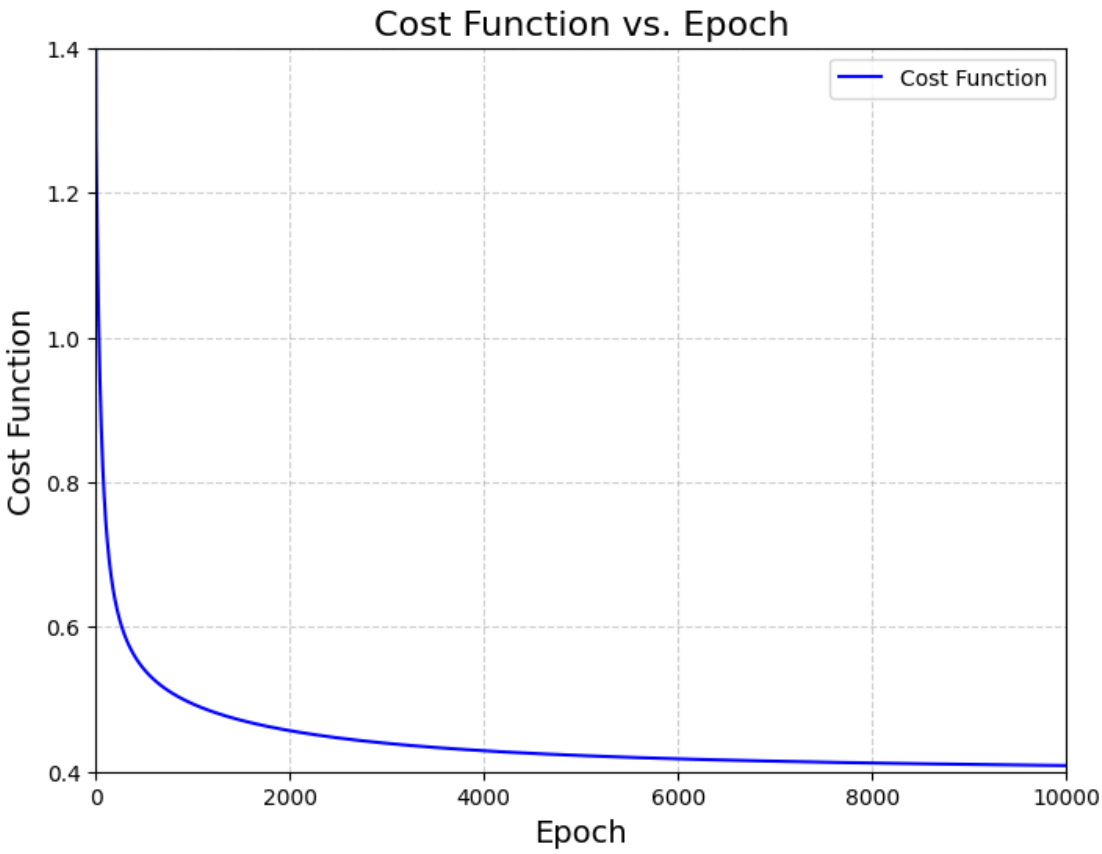

plt.title("Cost Function vs. Epoch", fontsize=16)
plt.xlabel("Epoch", fontsize=14)
plt.ylabel("Cost Function", fontsize=14)


plt.grid(True, linestyle="--", alpha=0.6)


plt.xlim(0, epoch)
plt.ylim(0.4, 1.4)


plt.legend()


plt.show()
########################
```



0.4 4. Plot result

4.1 Plot scatters using ground-truth label. (2 points)

[66]:
```
#########################
# Write your code here #
def plot_scatter(sepal_length, sepal_width, species):

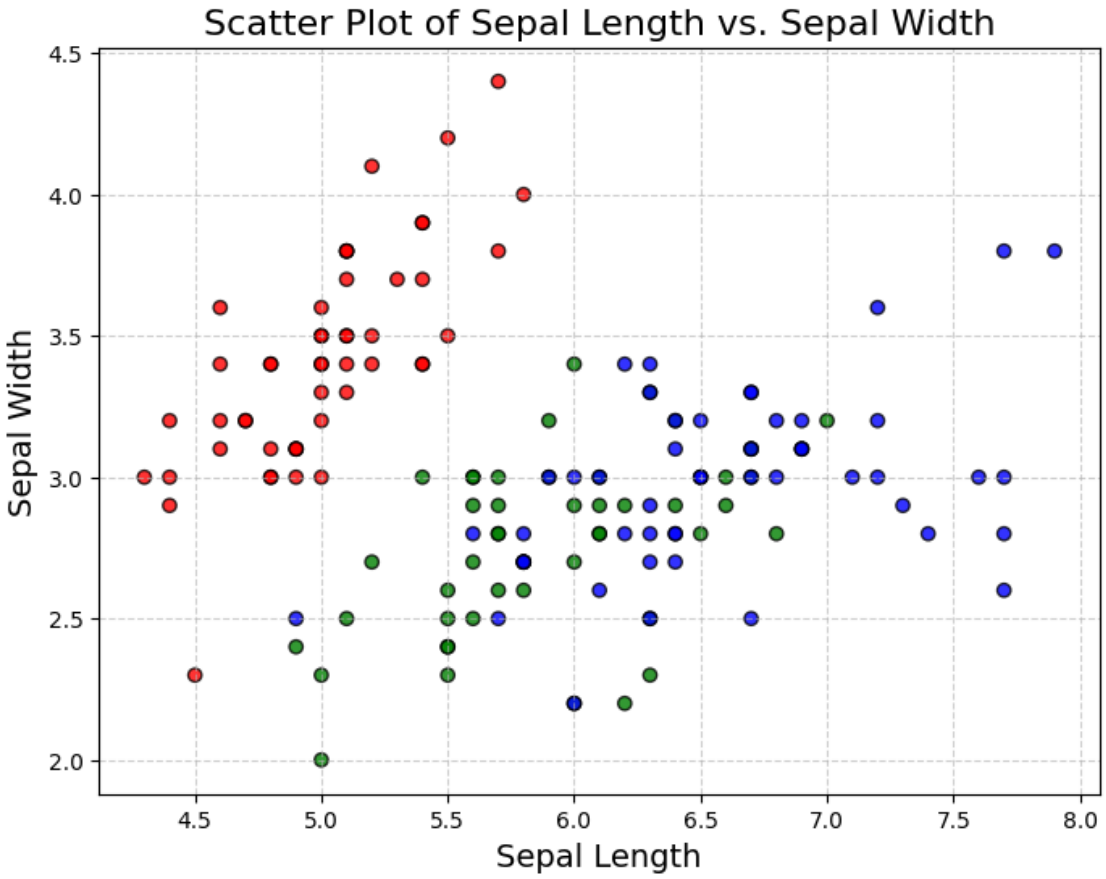
    color_map = {0: 'red', 1: 'green', 2: 'blue'}
    colors = [color_map[label] for label in species.ravel()]


    plt.figure(figsize=(8, 6))
    plt.scatter(sepal_length, sepal_width, c=colors, alpha=0.8,␣
 ↪edgecolors='black')

    plt.title("Scatter Plot of Sepal Length vs. Sepal Width", fontsize=16)
    plt.xlabel("Sepal Length", fontsize=14)
    plt.ylabel("Sepal Width", fontsize=14)
```

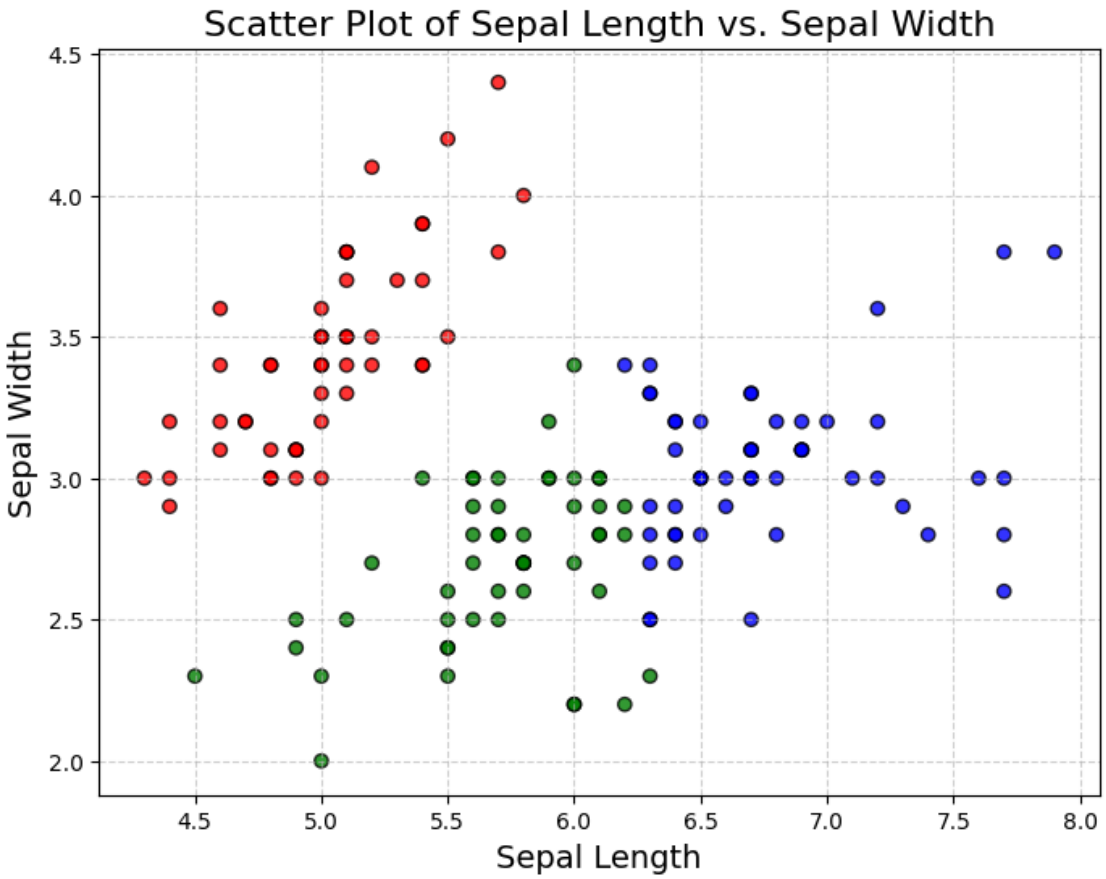
```
    plt.grid(True, linestyle="--", alpha=0.6)

    plt.show()
plot_scatter(sepal_array[:, 0], sepal_array[:, 1], species_array)
##########################
```



4.1 Plot scatters using predicted label. (2 points)

[67]:
```
##########################
# Write your code here #
def predict(X, w):
    logits = np.dot(X, w)
    y_pred = softmax(logits)
    predicted_labels = np.argmax(y_pred, axis=1).reshape(-1, 1)
    return predicted_labels
predicted_labels = predict(X, w)
plot_scatter(X[:, 1], X[:, 2], predicted_labels)
##########################
```


Scatter Plot of Sepal Length vs. Sepal Width
Sepal Width
Sepal Length
4.5
4.0
3.5
3.0
2.5
2.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0